

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Final Examination of Semester 1

Year: 2024

Subject: 031021111 Electrical Materials and Calculation

Section: 5-6

Date: 9 October 2024

Time: 10.00-12.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. There will be 8 pages (including this page) and they are worth 65 points.

- This exam paper contains no errors. If a suspected error is found, it is the student's discretion to correct it.
- Answer the questions on this test paper.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators can be used in this test.

Now begin the test.

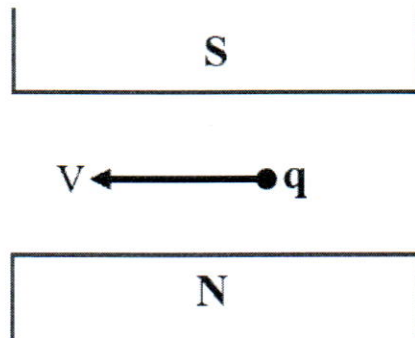
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Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

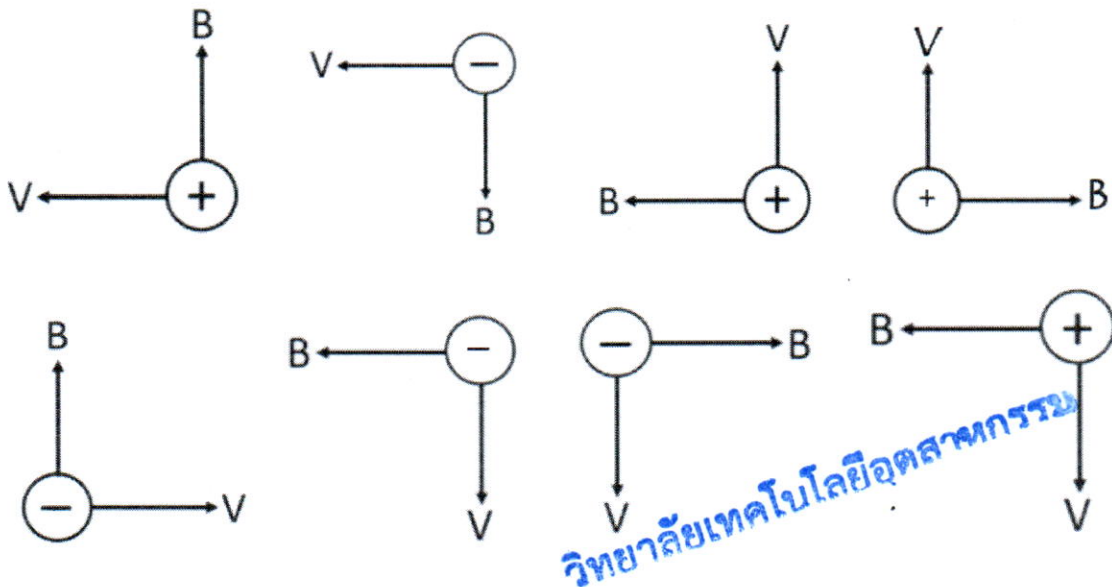
This exam paper is part of a set used to assess student abilities as follows:

		Q1-Q2, Q11	Q3, Q9-Q10	Q4-Q6	Q7-Q8
CLO 1	Calculate magnetic field correctly	✓			
CLO 2	Calculate electric a field correctly		✓		
CLO 3	Use semi-conductor, insulating oil and cable correctly			✓	
CLO 4	Calculate capacitance in electric circuit				✓

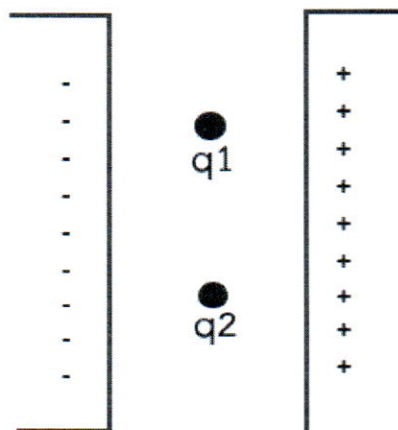
1. Draw the direction of “magnetic field (B)”, and “magnetic force (F_m)” of the proton (q) in the figure below. (2 points)



2. Draw the direction of the magnetic force of the figures below. Also, specify “in” or “out” on those force direction. Note: “in” means into the viewer, while “out” means out of the viewer. (8 points)



3. Draw the direction of “electric force (F_e) of proton (q_1)”, and “electric force (F_e) of electron (q_2)”, as shown in the following figure. (2 points)



4. Match the definition and function as written in Table 1 with the alphabet as follows.
(8 points)

- | | | |
|------------------|---------------------|----------------------|
| A. Armour | D. Filler | I. Conductor Shield |
| B. Bedding | E. Conductor screen | J. Individual screen |
| C. Thermosetting | H. Thermoplastic | |

Table 1: Definition and Function

Alphabet	Definition and Function
.....	To keep uniform electric field
.....	To keep electromagnetic radiation in
.....	To against bundle hit each other
.....	be soften on heating and harden again on cooling
.....	To protect noise/radiation
.....	Never soften again after harden once
.....	To keep bundle together
.....	To be mechanical protection of bundle

5. Fill in the effect of transformer oil temperature in the table below. (6 points)

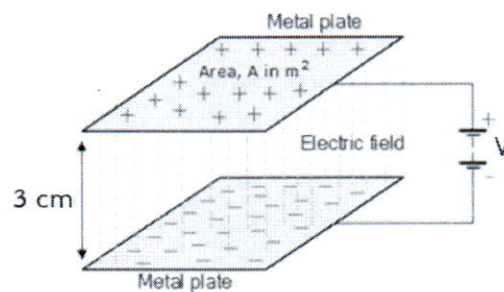
Oil temperature	Oil volume	Size of rubber bag	Flow of air
increasing			
decreasing			

6. Fill in the doping process and its effect of each semiconductor type. (6 points)

Semiconductor type	Doped by	Number of electrons	Resistor
P-type			
N-type			

Note: Please show step-by-step how to get the solutions for question 7-11. Both quantity and unit must be answered. (33 points)

7. The circuit of the capacitance is shown below. The dielectric between metal plates with a total surface area of 0.2 m^2 is made from mica ($\epsilon_r=5.40$). The charge is $1,600 \text{ pC}$. Assume that $\epsilon_0=8.84 \text{ pF/m}$. (6 points)



- 7.1 Calculate the amount of capacitance of this circuit. (2 points)

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- 7.2 Calculate the amount of voltage supplied to two plates. (2 points)

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7.3 Calculate the amount of energy in this capacitor. (2 points)

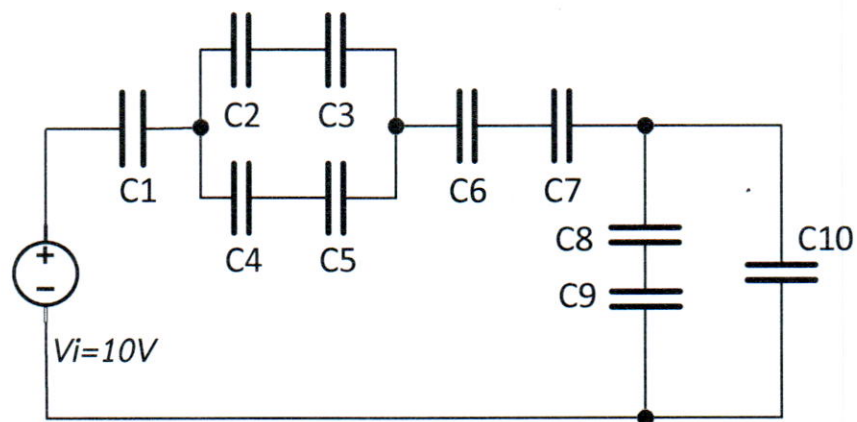
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8. Calculate the total capacitance of the following circuit. Assume that the value of each capacitor is $6\ \mu\text{F}$. (6 points)

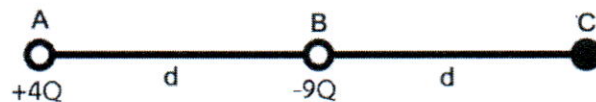


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9.2 Calculate the magnitude and draw the direction of the electric force when the charge of $+50 \mu\text{C}$ is placed at the “N” point. (2 points)

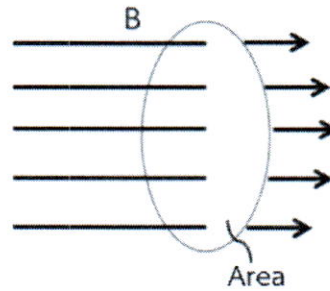
10. The electric field is created by the charge “A” and “B” at point “C”, as shown in the figure. Assume that there is a positive charge (+Q) at point “C”, $d=0.5\text{m}$, $Q=8\text{ nC}$ and $k=9\times 10^9\text{ N.m}^2/\text{C}^2$.

Calculate the magnitude and draw the direction of the electric field at point "C".
(6 points)



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11. The magnetic field lines pass through the sphere with the surface area of 5 m^2 and the magnetic field of $16,000 \text{ Tesla}$. The positive charge with $350 \text{ } \mu\text{C}$ moves perpendicular to the magnetic field from bottom to top of the sphere with speed of 2 m/s . (5 points)



- 11.1 Calculate the magnetic flux. (2 points)

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- 11.2 Calculate the magnitude and draw the direction of the magnetic force. (3 points)

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